

MAT 426, Class 07: Inner Products

Definition. If $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ are vectors in $\mathbb{R}^n$, then the **inner product** $\mathbf{u}$ and $\mathbf{v}$ is the
_____ $\mathbf{u} \cdot \mathbf{v} =$

Example 1. Calculate the inner product of $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 6 \\ 5 \end{bmatrix}$.

THEOREM Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^n$ and c be a scalar. Then:

- (a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- (b) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
- (c) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
- (d) $\mathbf{0} \cdot \mathbf{u} = 0$.

Proof of (2).

Proof of (3).

Definition. The length (or magnitude) $\|\mathbf{u}\|$ of a vector $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ is $\|\mathbf{u}\| =$

Example 2. Compute the length of the vector with initial point $(0, 1, 3)$ and terminal point $(-1, 3, 2)$.

Definition. A unit vector $\mathbf{u}$ is a vector whose _____ equals _____. In other words, $\mathbf{u} \cdot \mathbf{u} = \underline{\hspace{2cm}}$

For any non-zero vector $\mathbf{u}$, $\mathbf{u}/\|\mathbf{u}\|$ is the unit vector in the same direction as $\mathbf{u}$. Why?

Example 3. Find a unit vector in the direction of $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$.

Two Important Inequalities:

CAUCHY-SCHWARZ INEQUALITY For any two vectors $\mathbf{u}, \mathbf{v}$, we have $\|\mathbf{u} \cdot \mathbf{v}\| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$

TRIANGLE INEQUALITY For any two vectors $\mathbf{u}, \mathbf{v}$, $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Proof of Triangle Inequality:

Example 4. Consider two perpendicular lines in $\mathbb{R}^2$, one pointing in the direction of $\mathbf{u}$ and the other in the direction of a vector $\mathbf{v}$. We say that lines are perpendicular if the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u}$ to $-\mathbf{v}$. How can this characterization of perpendicular be captured using the dot product?

Definition. Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal if _____

Example 5. Let $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ -3 \end{bmatrix}$. Determine if $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

Definition.

- If a vector $\mathbf{z}$ is orthogonal to _____ vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be _____ to W .
- The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the _____ of W and is denoted by $W^\perp$

THEOREM

- A vector $\mathbf{x}$ is in $W^\perp$ if and only if $\mathbf{x}$ is orthogonal to every vector in a set that spans W .
- $W^\perp$ is a subspace of $\mathbb{R}^n$

Proof:

Definition. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in $\mathbb{R}^n$ is called an **orthogonal set** if each _____ of _____ vectors in the set are _____.

Example 6. Determine whether $\left\{ \begin{bmatrix} 4 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$ is an orthogonal set in $\mathbb{R}^3$.

Example 7. Determine whether $\left\{ \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthogonal set in $\mathbb{R}^3$.

THEOREM If $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent.

Proof.

Definition. An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a _____ for W that is an _____ set.

THEOREM Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$ and let $\mathbf{y}$ be any vector in W . Then, the weights $c_1, c_2, \dots, c_p$ in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

Proof.

Example 8. Show that $\left\{ \begin{bmatrix} 4 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$ form an orthogonal basis for $\mathbb{R}^3$ and express $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as a linear combination of these vectors.

Example 9. Show that $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -6 \\ 2 \end{bmatrix} \right\}$ form an orthogonal basis for $\mathbb{R}^2$ and express $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ as a linear combination of these vectors.

Definition. A set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an _____ set of _____ vectors. If W is the subspace spanned by such a set, then, $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an orthonormal _____ for W .

Example 10. Determine whether the given set of vectors $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthonormal set. If not, normalize the vectors to form an orthonormal set.

Example 11. Determine whether the given set of vectors $\left\{ \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} \right\}$ is orthonormal. If not, normalize the vectors to form an orthonormal set.